In[ ]:=
```
n1 = 8;
g1 = 3;
n2 = 4;
g2 = 2;
n3 = 4;
g3 = 2;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.5062272992248755;
β12 = 0.20906053042894954;
β22bar = 0.7841337886442463;

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.5307692307692308; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -1.3501437082633336;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
```

Q3 $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the middle region. This is obtained by solving two

consistency conditions along with other constant γ12bar. *)

```
Q2 = -3.64394796997568;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
```

Q2 $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the inner region. This is obtained by solving three

consistency conditions along with other constants β13 & β23. *)

```
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
```

$$R = \frac{3\,a}{(8 * alfas)} * \left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{2}{3}};$$(*Page number 54, it is inverse in the copy though *)

$$c = \frac{12}{5\,\pi}\,\frac{4}{3}\,g^2\,\frac{4}{3}\,\frac{alfas}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}};$$

(*Multiplication factor in the expression of kinetic energy *)

$$d = \frac{4}{3}\,g^2\,\frac{4}{3}\,\frac{alfas}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}}\,\frac{18}{25\,\pi}\,\frac{\eta}{\left(\eta-\frac{1}{5}\right)};\quad (*\ \frac{18g^2}{5(5\eta\ -1\ )}.16\ \pi^2.\left(\frac{3a}{8alfas}\right).\ \frac{R^2}{(6\pi^2)^2}\ ,$$

The value of R is from page number 54 which is written inverse here,

and $\left(\frac{3a}{8alfas}\right)$ is inverse in copy, BA, 13 September *)

(*Multiplication factor in the expression of Potential energy *)

(*Let us define the three functions for three regions *)

```
f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
```

```
Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
 (*Appendix from Kinetic energy*) /.
 First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)]/x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(* s2 is the first derivative of function f_2(x) *)
(*u2 is the function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
  (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)]/x^(1/2)),
    y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
    (* It means at the boundary x3 the function df1 changes to other function *)
    y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
    (*It means at the boundary x3 the function f1 changes to other function*)
   }, (* It means at the boundary x3 the function f1 changes to other function *)
   y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region which is solved by using boundary

condition of $\frac{d}{dx}f_1(x)\big|_{x_2}=\gamma 12bar\frac{d}{dx}f_2(x)\big|_{x_2}$ and $f_1(x3)=\gamma 12 f_2(x3)$ *)

```
f2[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2), y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3 / 2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2 / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).   *)

(*Start the program*)

m2 = 40;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
```

```
m2B = m2;
Energy = 10^9;
Energy = 10^9;
Energy2plus = 10^9;
H = 10^9 + 1;
SN = 0;
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)] / x^(1/2)),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];


  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2,  u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2,  u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];
```

```
x1 = x3 + 0.1;
FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
solution1 = NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
    y1'[x3] == γ12bar * s3 , y1[x3] == γ12bar * u3 }, y1, {x, x3, x1}];
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (* u1 is the function f_1(x) *)
s1 = t6[[1]]; (* s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)

normmx2 = x2 s2 - u2 ;
normx2 = (n2 Q2) / (3 η) ;
(*normx2barr and normx2bar should be equal *)
normmx3 = x3 s3 - u3 ;
normx3 = (n3 Q3) / (3 η) + β22bar (n2 Q2) / (3 η) - (β22bar)^(3/2) (n2 Q3) / (3 η);
(*normmx2 and normx2 should be equal *)
normmx1 = x1 s1 - u1 ;
normx1 = (n1 Q1) / (3 η) + γ12bar (n3 Q3) / (3 η) - (γ12bar)^(3/2) (n3 Q1) / (3 η) +
  ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) (n2 Q1) / (3 η) - γ12bar (β22bar)^(3/2) (n2 Q3) / (3 η) + γ12bar β22bar (n2 Q2) / (3 η) ;
(*normmx1 and normx1 should be equal *)

h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
```

```
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
h6 = h3[[1]] ;
```

$$AB = \frac{4}{7}\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{5\,n2}{21\,\eta}\,s2\,;$$

$$CD = \frac{4}{7}\sqrt{x3}\,(u3)^{\frac{5}{2}} - \frac{4}{7}(\beta 22bar)^{\frac{5}{2}}\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{5}{7\,Q3}\,s3\left(\frac{n3\,Q3}{3\,\eta} + \beta 22bar\,\frac{n2\,Q2}{3\,\eta} - (\beta 22bar)^{\frac{3}{2}}\,\frac{n2\,Q3}{3\,\eta}\right) + \frac{5}{7\,Q3}(\beta 22bar)^2\,\frac{n2\,Q2}{3\,\eta}\,s2\,;$$

$$CCD = \frac{4}{7}\sqrt{x3}\,(u3)^{\frac{5}{2}} - \frac{4}{7}\sqrt{x2}\,(\beta 22bar)^{\frac{5}{2}}(u2)^{\frac{5}{2}} - \frac{5}{7\,Q3}\,s3\,normmx3 + \frac{5}{7\,Q3}(\beta 22bar)^2\,normmx2\,s2\,;$$

$$EF = \frac{4}{7}\sqrt{x1}\,(u1)^{\frac{5}{2}} - \frac{4}{7}(\gamma 12bar)^{\frac{5}{2}}\sqrt{x3}\,(u3)^{\frac{5}{2}} - \frac{5}{7\,Q1}\,s1\left(\frac{n1\,Q1}{3\,\eta} + \gamma 12bar\,\frac{n3\,Q3}{3\,\eta} - (\gamma 12bar)^{\frac{3}{2}}\,\frac{n3\,Q1}{3\,\eta} + \frac{n2\,Q1}{3\,\eta}\left((\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - (\beta 12)^{\frac{3}{2}}\right) - (\gamma 12bar)(\beta 22bar)^{\frac{3}{2}}\,\frac{n2\,Q3}{3\,\eta} + (\gamma 12bar)(\beta 22bar)\,\frac{n2\,Q2}{3\,\eta}\right) + \frac{5}{7\,Q1}(\gamma 12bar)^2\,s3\left(\frac{n3\,Q3}{3\,\eta} + \beta 22bar\,\frac{n2\,Q2}{3\,\eta} - (\beta 22bar)^{\frac{3}{2}}\,\frac{n2\,Q3}{3\,\eta}\right);$$

```
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)
```

$$K222 = \frac{n2}{3\,\eta\,Q2}\,s2 - \frac{1}{Q2}\,h4\,;$$

$$K22 = \frac{4\,n2}{7\,\eta\,Q2}\,s2 - \frac{4}{7\,Q2}\sqrt{x2}\,(u2)^{\frac{5}{2}};$$

```
(*K2bar2barr and K2bar2bar should be equal *)
```

$$K32 = -\frac{4}{7\,Q2}\,\beta22bar^{1.5}\sqrt{x2}\,(u2)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2} - \frac{\beta22bar}{Q3} + \frac{5*\beta22bar^{1.5}}{7\,Q2}\right)s2 + \frac{n2}{3\,\eta\,Q3}\,s3\,;$$

$$K332 = \frac{n2\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}\,s2 - \frac{n2\,\beta22bar}{3\,\eta\,Q3}\,s2 + \frac{n2}{3\,\eta\,Q3}\,s3 - \frac{(\beta22bar)^{\frac{3}{2}}}{Q2}\,h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21 = -\frac{4}{7\,Q2}\,\beta12^{1.5}\sqrt{x2}\,(u2)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\,\beta12^{1.5} + \frac{\beta12^{1.5}}{Q2} - \frac{\gamma12bar^{1.5}*\beta22bar}{Q3}\right)s2 + \frac{n2}{3\,\eta}\left(\frac{\gamma12bar^{1.5}}{Q3} - \frac{\gamma12bar}{Q1}\right)s3 + \frac{n2}{3\,\eta\,Q1}\,s1\,;$$

$$K221 = \frac{n2}{3\,\eta\,Q2}\,(\beta12)^{\frac{3}{2}}\,s2 + \frac{n2\,(\gamma12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}\,s3 - \frac{n2}{3\,\eta\,Q3}\,(\gamma12bar)^{\frac{3}{2}}\,\beta22bar\,s2 + \frac{n2}{3\,\eta\,Q1}\,s1 - \frac{n2}{3\,\eta\,Q1}\,\gamma12bar\,s3 - \frac{(\beta12)^{\frac{3}{2}}}{Q2}\,h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31 = \frac{4}{7}\,\beta22bar^{1.5}\left(\frac{\gamma12bar^{1.5}*\beta22bar}{Q3} - \frac{\beta12^{1.5}}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{4}{7\,Q3}\,\gamma12bar^{1.5}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\,\beta22bar\left(\frac{5}{7\,Q2}\,\beta22bar^{\frac{1}{2}}*\beta12^{1.5} - \frac{\gamma12bar^{1.5}*\beta22bar^{1.5}}{Q3} - \frac{5\,Q2}{7\,Q3^2}\,\gamma12bar^{1.5}*\beta22bar + \frac{Q2}{Q3^2}\,\gamma12bar^{1.5}*\beta22bar + \frac{1}{Q2}\,\beta12^{1.5}*\beta22bar^{\frac{1}{2}} - \frac{\beta12^{1.5}}{Q3}\right)s2 + \frac{1}{3\,\eta\,Q3}\left(n3*\gamma12bar^{1.5} - \frac{n3\,Q3}{Q1}\,\gamma12bar + n2*\beta12^{1.5} - \frac{n2\,Q2}{Q3}\,\gamma12bar^{1.5}*\beta22bar +\right.$$

$$\frac{5}{7}\gamma 12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta 22bar-n2*\beta 22bar^{1.5}\right)\Big)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122=\frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\Big[\frac{1}{Q3}s3-\frac{\beta 22bar}{Q3}s2\Big]-\frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4-\frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$-\frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\Big[\frac{1}{Q3}s3-\frac{\beta 22bar}{Q3}s2\Big]+\frac{n3}{3\,\eta\,Q1}s1-\frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+(\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\Big[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2+\frac{1}{Q3}s3-\frac{\beta 22bar}{Q3}s2\Big]+$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3-\frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\;;\;*)$$

$$K331=\frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3-\frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2-\frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4-\frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5-$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3+\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2+\frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2}-\frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2+\frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3}-\frac{1}{Q1}\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

(*K12 and K122 should be equal*)

$$K33=\frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3}-\frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}}-$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2}+\frac{2}{7}\frac{Q2}{Q3^2}-\frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2+$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3}+\beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}}-\frac{2}{7}\frac{Q2}{Q3}-\frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\;;$$

$$K333=\frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}-\frac{\beta 22bar}{Q3}\right)s3-\frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}-\frac{\beta 22bar}{Q3}\right)s2-$$

$$\frac{(\beta 22bar)^3}{Q2}\,h4 - \frac{1}{Q3}\,h5 + (\beta 22bar)^3\,\frac{n2}{3\,\eta\,Q2}\,s2 - (\beta 22bar)^{\frac{5}{2}}\,\frac{n2}{3\,\eta\,Q3}\,s2 + \frac{n3}{3\,\eta\,Q3}\,s3\,;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta 22bar^{\frac{5}{2}} * \gamma 12bar^3}{Q3} - \frac{\beta 12^3}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma 12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma 12bar^3}{Q3}\right)\sqrt{x3}\,(u3)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q1}\sqrt{x1}\,(u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta 12^3}{Q2} - \frac{\beta 12^{\frac{3}{2}} * \gamma 12bar^{\frac{3}{2}} * \beta 22bar}{Q3} + \frac{Q2}{Q3^2}\,\gamma 12bar^3 * \right.\right.$$

$$\left.\beta 22bar^2 - \frac{\beta 12^{1.5} * \gamma 12bar^{1.5} * \beta 22bar}{Q3}\right) +$$

$$\left.\frac{5\,n2}{21\,\eta}\left(\frac{\beta 12^3}{Q2} - \frac{Q2}{Q3^2}\,\beta 22bar^2 * \gamma 12bar^3\right)\right)s2 +$$

$$\left(\frac{n2}{3\,\eta}\left(\frac{\beta 12^{1.5} * \gamma 12bar^{1.5}}{Q3} - \frac{\beta 12^{1.5} * \gamma 12bar}{Q1} - \frac{\beta 22bar^{1.5} * \gamma 12bar^3}{Q3} + \right.\right.$$

$$\frac{\beta 22bar^{1.5} * \gamma 12bar^{\frac{5}{2}}}{Q1} + \frac{\beta 12^{1.5} * \gamma 12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\,\gamma 12bar^3 * \beta 22bar -$$

$$\frac{\beta 12^{1.5} * \gamma 12bar}{Q1} + \frac{\gamma 12bar^{\frac{5}{2}} * \beta 22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\,\gamma 12bar^2 * \beta 22bar -$$

$$\left.\left.\frac{Q3}{Q1^2}\,\beta 22bar^{1.5} * \gamma 12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma 12bar^3}{Q3} - \frac{2 * \gamma 12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\,\gamma 12bar^2\right) +\right.$$

$$\frac{5}{21\,\eta}\left(\frac{n3}{Q3}\,\gamma 12bar^3 + \frac{n2\;Q2}{Q3^2}\,\gamma 12bar^3 * \beta 22bar - \frac{n2}{Q3}\,\beta 22bar^{1.5} * \gamma 12bar^3 -\right.$$

$$\left.\left.\frac{n3\ Q3}{Q1^2}\,\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\,\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\,\gamma12bar^2 * \beta22bar^{1.5}\right)\right) *$$

$$s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\,\gamma12bar + \frac{n3}{3\,\eta\,Q1}\,\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}\right.$$

$$\left(\beta12^{1.5} - \gamma12bar^{1.5} * \beta22bar^{1.5} - \frac{Q2}{Q1}\,\beta22bar * \gamma12bar + \frac{Q3}{Q1}\,\beta22bar^{1.5} * \gamma12bar\right) +$$

$$\frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\,\gamma12bar - \frac{n3}{Q1}\,\gamma12bar^{1.5} + \frac{n2}{Q1}\,\gamma12bar^{1.5} * \beta22bar^{1.5} -\right.$$

$$\left.\left.\frac{n2}{Q1}\,\beta12^{1.5} - \frac{n2\ Q3}{Q1^2}\,\beta22bar^{1.5} * \gamma12bar + \frac{n2\ Q2}{Q1^2}\,\gamma12bar\ \beta22bar\right)\right) * s1\ ;$$

$$K111 = \frac{-(\beta12)^3}{Q2}\,h4 - \frac{(\gamma12bar)^3}{Q3}\,h5 - \frac{1}{Q1}\,h6 + \frac{n2}{3\,\eta\,Q3}\,(\beta12)^{\frac{3}{2}}\,(\gamma12bar)^{\frac{3}{2}}\,s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}\,(\gamma12bar)^3$$

$$\beta22bar\,s3 - \frac{n2}{3\,\eta\,Q3}\,(\beta12)^{\frac{3}{2}}\,(\gamma12bar)^{\frac{3}{2}}\,\beta22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}\,(\gamma12bar)^3\,(\beta22bar)^2\,s2$$

$$+ \left(\frac{n2}{3\,\eta\,Q1}\,(\beta12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}\,(\gamma12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\,\gamma12bar - \frac{n2}{3\,\eta\,Q1}\,(\gamma12bar)^{\frac{3}{2}}\,(\beta22bar)^{\frac{3}{2}} -\right.$$

$$\left.\frac{n2\,Q2}{3\,\eta\,Q1^2}\,\beta22bar\,\gamma12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}\,(\beta22bar)^{\frac{3}{2}}\,\gamma12bar\right)s1 -$$

$$\left(\frac{n2}{3\,\eta\,Q1}\,(\beta12)^{\frac{3}{2}}\,\gamma12bar + \frac{n3}{3\,\eta\,Q1}\,(\gamma12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\,(\gamma12bar)^2 - \frac{n2}{3\,\eta\,Q1}\,(\gamma12bar)^{\frac{5}{2}}\right.$$

$$\left.(\beta22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\,(\gamma12bar)^2\,\beta22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}\,(\beta22bar)^{\frac{3}{2}}\,(\gamma12bar)^2\right)s3 +$$

$$\frac{n2}{3\,\eta}\,\frac{(\beta12)^3}{Q2}\,s2 - \frac{n2}{3\,\eta\,Q3}\,(\beta12)^{\frac{3}{2}}\,(\gamma12bar)^{\frac{3}{2}}\,\beta22bar\,s2 + \frac{n2}{3\,\eta\,Q3}\,(\beta12)^{\frac{3}{2}}\,(\gamma12bar)^{\frac{3}{2}}\,s3 -$$

$$\frac{n2}{3\,\eta\,Q1}\,(\beta12)^{\frac{3}{2}}\,\gamma12bar\,s3 + \frac{n3}{3\,\eta\,Q3}\,(\gamma12bar)^3\,s3 - \frac{n2}{3\,\eta\,Q3}\,(\beta22bar)^{\frac{3}{2}}\,(\gamma12bar)^3\,s3 -$$

$$\frac{n3}{3\,\eta\,Q1}\,(\gamma12bar)^{\frac{5}{2}}\,s3 + \frac{n2}{3\,\eta\,Q1}\,(\beta22bar)^{\frac{3}{2}}\,(\gamma12bar)^{\frac{5}{2}}\,s3 + \frac{n1}{3\,\eta\,Q1}\,s1\ ;$$

```
(*K11 and K111 should be equal *)
```

$$T1 = c\left(\frac{4}{7}\left(\text{alfa1}*\text{g1}\left(\beta 12^{\frac{5}{2}} - \gamma 12\text{bar}^{\frac{5}{2}}*\beta 22\text{bar}^{\frac{5}{2}}\right) + \text{alfa2}*\text{g2}\right)\sqrt{\text{x2}}\ (\text{u2})^{\frac{5}{2}} + \right.$$

$$\frac{4}{7}\text{alfa2}*\text{g3}\sqrt{\text{x3}}\ (\text{u3})^{\frac{5}{2}} + \frac{4}{7}\text{alfa1}*\text{g1}\sqrt{\text{x1}}\ (\text{u1})^{\frac{5}{2}} +$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^2\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}} + \text{alfa2}*\text{g3}\right) - \right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}} - \text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}} - \text{alfa2}*\text{g2}\right)\text{s2} +$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^2*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3} + \beta 22\text{bar}*\text{n2}*\text{Q2} - \beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right) - \right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}} + \text{alfa2}*\text{g3}\right)\left(\text{n3} + \beta 22\text{bar}\frac{\text{n2}*\text{Q2}}{\text{Q3}} - \beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3} -$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1} + \gamma 12\text{bar}\frac{\text{n3}\,\text{Q3}}{\text{Q1}} + \gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2} - \text{n3}\right) - \right.$$

$$\left.\beta 12^{1.5}*\text{n2} - \gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\frac{\text{n2}\,\text{Q3}}{\text{Q1}} + \gamma 12\text{bar}\,\beta 22\text{bar}\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$V1 = -\text{d}\left(\frac{(\text{g1}\,\text{n1} - 1)}{\text{n1}}\text{g1}\,\text{K11} + \frac{(\text{g3}\,\text{n3} - 1)}{\text{n3}}\text{g3}\,\text{K33} + \right.$$

$$\left.\frac{(\text{g2}\,\text{n2} - 1)}{\text{n2}}\text{g2}\,\text{K22} + 2\ \text{g1}\,\text{g3}\,\text{K31} + \frac{4}{3}\text{g2}\,\text{g1}\,\text{K21} + \frac{4}{3}\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
```

```
    Print["normx3 (computationally)=  ", normmx3];
    Print["normx3 (by hand)=  ", normx3];
    Print["normx1 (computationally)=  ", normmx1];
    Print["normx1 (by hand)=  ", normx1];
    Print["0 to x2 (h4 computationally)=  ", h4];
    Print["0 to x2 (h4 by hand)=  ", AB];
    Print["x2 to x3 (h5 computationally)=  ", h5];
    Print["x2 to x3 (h5 by hand)=  ", CD];
    Print["x2 to x3 (h5 by handddddd)=  ", CCD];
    Print["x3 to x1 (h6 computationally)=  ", h6];
    Print["x3 to x1 (h6 by hand)=  ", EF];
    Print["K22  (computationally) = ", K222];
    Print["K22  (by hand) = ", K22];
    Print["K32  (computationally) = ", K332];
    Print["K32  (by hand) = ", K32];
    Print["K21  (computationally) = ", K221];
    Print["K21  (by hand) =", K21];
    Print["K31  (computationally) = ", K331];
    Print["K31  (by hand) = ", K31];
    Print["K33  (computationally) = ", K333];
    Print["K33  (by hand) = ", K33];
    Print["K11  (computationally) = ", K111];
    Print["K11  (by hand) = ", K11];
    Print["x2= ", x2];
    Print["x3= ", x3];
    Print["x1=  ", x1];
     Print["T by hand=", T1];
    Print["V by hand=", V1];
    Print["m2=", m2];
    Print["R=  ", R];
Print["Energywithoutmass = ", Energywithoutmass];
    Print["Energy=  ", Energy];
    If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
      Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
       H = 10^9 + 1;
       m2 = m2B;
       If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
   If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
   If[n == 3, Energy6bare = HX;
    Print["x1=", x1];
    Print["x3=", x3];
    Print["x2=", x2];
```

```
    Print["Kinetic Energy=", T1];
    Print["Potential Energy=", V1];
    Print["Volume Energy=", Volumeenergy];
   ]]];

(*Code for Plotting function *)
Plot1 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar* y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12* y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar* y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.0398218246
normx2 (computationally)=  -0.607324503
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.54499988
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.42306374
0 to x2 (h4 by hand)=  6.42306317
x2 to x3 (h5 computationally)=  1.48742413
x2 to x3 (h5 by hand)=  1.48742606
x2 to x3 (h5 by handdddd)=  1.48742473
x3 to x1 (h6 computationally)=  4.18197809
x3 to x1 (h6 by hand)=  4.18197891
K22  (computationally) = 0.262103861
K22  (by hand) = 0.262103706
K32  (computationally) = 0.244114294
K32  (by hand) = 0.244114187
K21  (computationally) = 0.280133092
K21  (by hand) =0.280133077
K31  (computationally) = 0.278270097
K31  (by hand) = 0.278270602
K33  (computationally) = 0.231224935
K33  (by hand) = 0.231226292
K11  (computationally) = 0.481556711
K11  (by hand) = 0.481558442
x2= 0.12787462
x3= 0.145173326
x1=  0.272469816
T by hand=1052.78385
```

```
V by hand=-226.662454
m2=40
R=  0.0398218246
Energywithoutmass = 1543.34794
Energy=  33735.3479
R=  0.0398218246
normx2 (computationally)=  -0.6073246
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999875
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.42470118
0 to x2 (h4 by hand)=  6.42470124
x2 to x3 (h5 computationally)=  1.48780815
x2 to x3 (h5 by hand)=  1.48780896
x2 to x3 (h5 by handdddd)=  1.48780875
x3 to x1 (h6 computationally)=  4.18309554
x3 to x1 (h6 by hand)=  4.18309641
K22  (computationally) = 0.262137382
K22  (by hand) = 0.2621374
K32  (computationally) = 0.244145434
K32  (by hand) = 0.244145447
K21  (computationally) = 0.280169588
K21  (by hand) =0.28016959
K31  (computationally) = 0.278305936
K31  (by hand) = 0.278306153
K33  (computationally) = 0.231253229
K33  (by hand) = 0.231253838
K11  (computationally) = 0.481619996
K11  (by hand) = 0.481621717
x2= 0.127858335
x3= 0.145154713
x1=  0.272433629
T by hand=1053.06136
V by hand=-226.691679
```

```
m2=40.01
R=  0.0398218246
Energywithoutmass = 1543.3105
Energy=  33 735.3105
R=  0.0398218246
normx2 (computationally)=  −0.607324601
normx2 (by hand)=  −0.607324662
normx3 (computationally)=  −0.544999875
normx3 (by hand)=  −0.544999908
normx1 (computationally)=  −0.434622675
normx1 (by hand)=  −0.434622687
0 to x2 (h4 computationally)=  6.43942922
0 to x2 (h4 by hand)=  6.43942929
x2 to x3 (h5 computationally)=  1.49128046
x2 to x3 (h5 by hand)=  1.49128127
x2 to x3 (h5 by handdddd)=  1.49128106
x3 to x1 (h6 computationally)=  4.19315227
x3 to x1 (h6 by hand)=  4.19315315
K22  (computationally) = 0.262436073
K22  (by hand) = 0.262436091
K32  (computationally) = 0.244424229
K32  (by hand) = 0.244424242
K21  (computationally) = 0.280497711
K21  (by hand) =0.280497712
K31  (computationally) = 0.278631998
K31  (by hand) = 0.278632215
K33  (computationally) = 0.231517769
K33  (by hand) = 0.231518378
K11  (computationally) = 0.48218929
K11  (by hand) = 0.482191015
x2= 0.127711983
x3= 0.144987632
x1=  0.272108607
T by hand=1055.55969
V by hand=−226.956415
m2=40.1
```

```
R=  0.0398218246
Energywithoutmass = 1542.98115
Energy=  33734.9812
R=  0.0398218246
normx2 (computationally)=  -0.607324581
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999897
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.45579387
0 to x2 (h4 by hand)=  6.45579381
x2 to x3 (h5 computationally)=  1.49513956
x2 to x3 (h5 by hand)=  1.49514089
x2 to x3 (h5 by handdddd)=  1.49514017
x3 to x1 (h6 computationally)=  4.20432693
x3 to x1 (h6 by hand)=  4.20432761
K22  (computationally) = 0.262767496
K22  (by hand) = 0.26276748
K32  (computationally) = 0.244733624
K32  (by hand) = 0.244733613
K21  (computationally) = 0.280861846
K21  (by hand) =0.280861844
K31  (computationally) = 0.278994045
K31  (by hand) = 0.278994397
K33  (computationally) = 0.231811916
K33  (by hand) = 0.231812889
K11  (computationally) = 0.482820634
K11  (by hand) = 0.482822038
x2= 0.127549953
x3= 0.144802663
x1=  0.271748805
T by hand=1058.33575
V by hand=-227.250213
m2=40.2
R=  0.0398218246
```

```
Energywithoutmass = 1542.63335
Energy=  33 734.6333
R=  0.0398218246
normx2 (computationally)=  −0.6073246
normx2 (by hand)=  −0.607324662
normx3 (computationally)=  −0.544999886
normx3 (by hand)=  −0.544999908
normx1 (computationally)=  −0.434622675
normx1 (by hand)=  −0.434622687
0 to x2 (h4 computationally)=  6.47215929
0 to x2 (h4 by hand)=  6.47215935
x2 to x3 (h5 computationally)=  1.49899746
x2 to x3 (h5 by hand)=  1.49899844
x2 to x3 (h5 by handdddd)=  1.49899807
x3 to x1 (h6 computationally)=  4.21550292
x3 to x1 (h6 by hand)=  4.2155037
K22  (computationally) = 0.263098627
K22  (by hand) = 0.263098644
K32  (computationally) = 0.245042662
K32  (by hand) = 0.245042673
K21  (computationally) = 0.281225546
K21  (by hand) =0.281225547
K31  (computationally) = 0.279355328
K31  (by hand) = 0.27935559
K33  (computationally) = 0.232104766
K33  (by hand) = 0.232105498
K11  (computationally) = 0.483451857
K11  (by hand) = 0.483453423
x2= 0.127388538
x3= 0.144618382
x1=  0.271390407
T by hand=1061.11201
V by hand=−227.543633
m2=40.3
R=  0.0398218246
Energywithoutmass = 1542.30461
```

```
Energy=  33734.3046
R=  0.0398218246
normx2 (computationally)=  -0.607324514
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999908
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.48852399
0 to x2 (h4 by hand)=  6.4885235
x2 to x3 (h5 computationally)=  1.50285803
x2 to x3 (h5 by hand)=  1.50286023
x2 to x3 (h5 by handdddd)=  1.50285864
x3 to x1 (h6 computationally)=  4.22667931
x3 to x1 (h6 by hand)=  4.2266799
K22  (computationally) = 0.263429014
K22  (by hand) = 0.263428878
K32  (computationally) = 0.245351185
K32  (by hand) = 0.245351091
K21  (computationally) = 0.281588765
K21  (by hand) =0.281588752
K31  (computationally) = 0.279716735
K31  (by hand) = 0.279717313
K33  (computationally) = 0.232398879
K33  (by hand) = 0.232400443
K11  (computationally) = 0.484081726
K11  (by hand) = 0.484083047
x2= 0.127227722
x3= 0.144434818
x1=  0.271033396
T by hand=1063.88847
V by hand=-227.836852
m2=40.4
R=  0.0398218246
Energywithoutmass = 1541.99453
Energy=  33733.9945
```

```
R=  0.0398218246
normx2 (computationally)=  -0.607324597
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999885
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.50489082
0 to x2 (h4 by hand)=  6.50489087
x2 to x3 (h5 computationally)=  1.5067149
x2 to x3 (h5 by hand)=  1.50671591
x2 to x3 (h5 by handdddd)=  1.50671552
x3 to x1 (h6 computationally)=  4.23785718
x3 to x1 (h6 by hand)=  4.23785797
K22  (computationally) = 0.263759532
K22  (by hand) = 0.263759544
K32  (computationally) = 0.245659547
K32  (by hand) = 0.245659555
K21  (computationally) = 0.281951574
K21  (by hand) =0.281951576
K31  (computationally) = 0.280076807
K31  (by hand) = 0.280077075
K33  (computationally) = 0.232690138
K33  (by hand) = 0.232690885
K11  (computationally) = 0.484711536
K11  (by hand) = 0.484713129
x2= 0.127067522
x3= 0.144251916
x1=  0.270677769
T by hand=1066.66504
V by hand=-228.12942
m2=40.5
R=  0.0398218246
Energywithoutmass = 1541.70334
Energy=  33733.7033
R=  0.0398218246
```

```
normx2 (computationally)=  -0.607324499
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999883
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.52125614
0 to x2 (h4 by hand)=  6.52125555
x2 to x3 (h5 computationally)=  1.51057551
x2 to x3 (h5 by hand)=  1.51057756
x2 to x3 (h5 by handdddd)=  1.51057613
x3 to x1 (h6 computationally)=  4.2490356
x3 to x1 (h6 by hand)=  4.2490364
K22  (computationally) = 0.264089062
K22  (by hand) = 0.264088899
K32  (computationally) = 0.245967268
K32  (by hand) = 0.245967155
K21  (computationally) = 0.282313902
K21  (by hand) =0.282313886
K31  (computationally) = 0.280437257
K31  (by hand) = 0.280437792
K33  (computationally) = 0.232983292
K33  (by hand) = 0.232984728
K11  (computationally) = 0.485340353
K11  (by hand) = 0.485342073
x2= 0.12690791
x3= 0.144069728
x1=  0.270323516
T by hand=1069.4419
V by hand=-228.421992
m2=40.6
R=  0.0398218246
Energywithoutmass = 1541.4304
Energy=  33733.4304
R=  0.0398218246
normx2 (computationally)=  -0.607324536
```

```
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999887
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.53762324
0 to x2 (h4 by hand)=  6.53762289
x2 to x3 (h5 computationally)=  1.51443408
x2 to x3 (h5 by hand)=  1.51443578
x2 to x3 (h5 by handdddd)=  1.5144347
x3 to x1 (h6 computationally)=  4.26021479
x3 to x1 (h6 by hand)=  4.26021557
K22  (computationally) = 0.26441862
K22  (by hand) = 0.264418524
K32  (computationally) = 0.246274823
K32  (by hand) = 0.246274756
K21  (computationally) = 0.282675796
K21  (by hand) =0.282675787
K31  (computationally) = 0.280796748
K31  (by hand) = 0.280797194
K33  (computationally) = 0.233274734
K33  (by hand) = 0.233275946
K11  (computationally) = 0.485968098
K11  (by hand) = 0.485969726
x2= 0.126748903
x3= 0.143888209
x1=  0.269970622
T by hand=1072.2188
V by hand=-228.713883
m2=40.7
R=  0.0398218246
Energywithoutmass = 1541.17593
Energy=  33733.1759
R=  0.0398218246
normx2 (computationally)=  -0.607324596
normx2 (by hand)=  -0.607324662
```

```
normx3 (computationally)=  -0.54499989
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622676
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.55399094
0 to x2 (h4 by hand)=  6.55399099
x2 to x3 (h5 computationally)=  1.51829242
x2 to x3 (h5 by hand)=  1.51829351
x2 to x3 (h5 by handdddd)=  1.51829304
x3 to x1 (h6 computationally)=  4.27139488
x3 to x1 (h6 by hand)=  4.27139565
K22  (computationally) = 0.264747848
K22  (by hand) = 0.26474786
K32  (computationally) = 0.246582035
K32  (by hand) = 0.246582043
K21  (computationally) = 0.283037249
K21  (by hand) =0.28303725
K31  (computationally) = 0.281155702
K31  (by hand) = 0.281155996
K33  (computationally) = 0.233565565
K33  (by hand) = 0.233566383
K11  (computationally) = 0.486595108
K11  (by hand) = 0.486596668
x2= 0.126590491
x3= 0.143707366
x1=  0.269619082
T by hand=1074.99586
V by hand=-229.005377
m2=40.8
R=  0.0398218246
Energywithoutmass = 1540.93965
Energy=  33732.9396
R=  0.0398218246
normx2 (computationally)=  -0.607324497
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999876
```

```
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622676
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.57035734
0 to x2 (h4 by hand)=  6.57035673
x2 to x3 (h5 computationally)=  1.52215329
x2 to x3 (h5 by hand)=  1.52215528
x2 to x3 (h5 by handdddd)=  1.52215391
x3 to x1 (h6 computationally)=  4.28257601
x3 to x1 (h6 by hand)=  4.28257691
K22  (computationally) = 0.265076155
K22  (by hand) = 0.265075987
K32  (computationally) = 0.246888609
K32  (by hand) = 0.246888492
K21  (computationally) = 0.283398243
K21  (by hand) =0.283398227
K31  (computationally) = 0.28151477
K31  (by hand) = 0.281515291
K33  (computationally) = 0.233857468
K33  (by hand) = 0.233858864
K11  (computationally) = 0.487221873
K11  (by hand) = 0.487223756
x2= 0.126432656
x3= 0.143527212
x1=  0.269268896
T by hand=1077.77326
V by hand=-229.296903
m2=40.9
R=  0.0398218246
Energywithoutmass = 1540.7212
Energy=  33 732.7212
R=  0.0398218246
normx2 (computationally)=  -0.607324599
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999908
normx3 (by hand)=  -0.544999908
```

```
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.58672623
0 to x2 (h4 by hand)=  6.58672629
x2 to x3 (h5 computationally)=  1.52601182
x2 to x3 (h5 by hand)=  1.52601313
x2 to x3 (h5 by handdddd)=  1.52601245
x3 to x1 (h6 computationally)=  4.29375736
x3 to x1 (h6 by hand)=  4.29375797
K22  (computationally) = 0.265404718
K22  (by hand) = 0.265404735
K32  (computationally) = 0.24719516
K32  (by hand) = 0.247195172
K21  (computationally) = 0.283758801
K21  (by hand) =0.283758802
K31  (computationally) = 0.281872805
K31  (by hand) = 0.281873155
K33  (computationally) = 0.234147574
K33  (by hand) = 0.234148551
K11  (computationally) = 0.487846607
K11  (by hand) = 0.487847896
x2= 0.126275421
x3= 0.143347717
x1=  0.268920033
T by hand=1080.55056
V by hand=-229.587502
m2=41.
R=  0.0398218246
Energywithoutmass = 1540.52078
Energy=  33732.5208
R=  0.0398218246
normx2 (computationally)=  -0.607324597
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999894
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
```

```
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.60309438
0 to x2 (h4 by hand)=  6.60309443
x2 to x3 (h5 computationally)=  1.52987136
x2 to x3 (h5 by hand)=  1.52987252
x2 to x3 (h5 by handdddd)=  1.52987199
x3 to x1 (h6 computationally)=  4.30494015
x3 to x1 (h6 by hand)=  4.30494088
K22  (computationally) = 0.265732539
K22  (by hand) = 0.265732552
K32  (computationally) = 0.247501137
K32  (by hand) = 0.247501146
K21  (computationally) = 0.284118922
K21  (by hand) =0.284118924
K31  (computationally) = 0.282230607
K31  (by hand) = 0.282230916
K33  (computationally) = 0.23443774
K33  (by hand) = 0.234438602
K11  (computationally) = 0.488471671
K11  (by hand) = 0.488473162
x2= 0.12611876
x3= 0.14316889
x1=  0.26857251
T by hand=1083.3282
V by hand=-229.878085
m2=41.1
R=  0.0398218246
Energywithoutmass = 1540.33797
Energy=  33732.338
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.0398218246
normx2 (computationally)=  -0.607324498
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999874
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.61946186
0 to x2 (h4 by hand)=  6.61946125
x2 to x3 (h5 computationally)=  1.5337326
x2 to x3 (h5 by hand)=  1.53373457
x2 to x3 (h5 by handdddd)=  1.53373323
x3 to x1 (h6 computationally)=  4.31612387
x3 to x1 (h6 by hand)=  4.31612478
K22  (computationally) = 0.266059644
K22  (by hand) = 0.266059476
K32  (computationally) = 0.247806583
K32  (by hand) = 0.247806466
K21  (computationally) = 0.284478594
K21  (by hand) =0.284478578
K31  (computationally) = 0.282588331
K31  (by hand) = 0.282588847
K33  (computationally) = 0.234728486
K33  (by hand) = 0.234729868
K11  (computationally) = 0.489096256
K11  (by hand) = 0.489098159
x2= 0.125962671
x3= 0.142990735
x1=  0.268226313
T by hand=1086.10611
V by hand=-230.168554
m2=41.2
R=  0.0398218246
Energywithoutmass = 1540.17258
Energy=  33732.1726
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.0398218246
normx2 (computationally)=  -0.607324542
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999874
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622674
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.63583123
0 to x2 (h4 by hand)=  6.63583091
x2 to x3 (h5 computationally)=  1.53759193
x2 to x3 (h5 by hand)=  1.53759342
x2 to x3 (h5 by handdddd)=  1.53759257
x3 to x1 (h6 computationally)=  4.32730815
x3 to x1 (h6 by hand)=  4.32730906
K22  (computationally) = 0.266386822
K22  (by hand) = 0.266386735
K32  (computationally) = 0.2481119
K32  (by hand) = 0.24811184
K21  (computationally) = 0.284837839
K21  (by hand) =0.28483783
K31  (computationally) = 0.282945135
K31  (by hand) = 0.282945527
K33  (computationally) = 0.235017651
K33  (by hand) = 0.235018713
K11  (computationally) = 0.489719445
K11  (by hand) = 0.489721294
x2= 0.125807166
x3= 0.142813227
x1=  0.267881421
T by hand=1088.88402
V by hand=-230.45829
m2=41.3
R=  0.0398218246
```

```
Energywithoutmass = 1540.02472
Energy=  33 732.0247
R=  0.0398218246
normx2 (computationally)=  −0.607324575
normx2 (by hand)=  −0.607324662
normx3 (computationally)=  −0.544999874
normx3 (by hand)=  −0.544999908
normx1 (computationally)=  −0.434622674
normx1 (by hand)=  −0.434622687
0 to x2 (h4 computationally)=  6.65220083
0 to x2 (h4 by hand)=  6.65220073
x2 to x3 (h5 computationally)=  1.54145164
x2 to x3 (h5 by hand)=  1.54145277
x2 to x3 (h5 by handdddd)=  1.54145228
x3 to x1 (h6 computationally)=  4.33849326
x3 to x1 (h6 by hand)=  4.33849416
K22  (computationally) = 0.266713565
K22  (by hand) = 0.266713538
K32  (computationally) = 0.24841683
K32  (by hand) = 0.248416811
K21  (computationally) = 0.285196645
K21  (by hand) =0.285196643
K31  (computationally) = 0.283301553
K31  (by hand) = 0.283301854
K33  (computationally) = 0.235306593
K33  (by hand) = 0.23530742
K11  (computationally) = 0.490341894
K11  (by hand) = 0.490343704
x2= 0.125652232
x3= 0.142636374
x1=  0.267537833
T by hand=1091.6621
V by hand=−230.747703
m2=41.4
R=  0.0398218246
Energywithoutmass = 1539.89407
```

```
Energy=  33731.8941
R=  0.0398218246
normx2 (computationally)=  -0.607324496
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999874
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622674
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.66856961
0 to x2 (h4 by hand)=  6.66856898
x2 to x3 (h5 computationally)=  1.54531347
x2 to x3 (h5 by hand)=  1.54531549
x2 to x3 (h5 by handdddd)=  1.54531411
x3 to x1 (h6 computationally)=  4.34967918
x3 to x1 (h6 by hand)=  4.34968009
K22  (computationally) = 0.267039547
K22  (by hand) = 0.267039375
K32  (computationally) = 0.248721216
K32  (by hand) = 0.248721096
K21  (computationally) = 0.285555002
K21  (by hand) =0.285554985
K31  (computationally) = 0.283657997
K31  (by hand) = 0.283658524
K33  (computationally) = 0.235596402
K33  (by hand) = 0.235597814
K11  (computationally) = 0.490963746
K11  (by hand) = 0.490965635
x2= 0.125497857
x3= 0.142460185
x1=  0.267195545
T by hand=1094.44046
V by hand=-231.037027
m2=41.5
R=  0.0398218246
Energywithoutmass = 1539.78037
Energy=  33731.7804
```

```
R=  0.0398218246
normx2 (computationally)=  -0.607324496
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999896
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622674
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.68493958
0 to x2 (h4 by hand)=  6.68493894
x2 to x3 (h5 computationally)=  1.54917456
x2 to x3 (h5 by hand)=  1.5491769
x2 to x3 (h5 by handdddd)=  1.54917521
x3 to x1 (h6 computationally)=  4.36086564
x3 to x1 (h6 by hand)=  4.36086634
K22  (computationally) = 0.267365395
K22  (by hand) = 0.267365222
K32  (computationally) = 0.249025377
K32  (by hand) = 0.249025257
K21  (computationally) = 0.285912927
K21  (by hand) =0.28591291
K31  (computationally) = 0.284013796
K31  (by hand) = 0.284014407
K33  (computationally) = 0.235885358
K33  (by hand) = 0.235887003
K11  (computationally) = 0.491584273
K11  (by hand) = 0.491585824
x2= 0.125344053
x3= 0.142284638
x1=  0.266854539
T by hand=1097.21886
V by hand=-231.325773
m2=41.6
R=  0.0398218246
Energywithoutmass = 1539.68368
Energy=  33731.6837
R=  0.0398218246
```

```
normx2 (computationally)=  -0.607324586
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.54499988
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622674
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.70131088
0 to x2 (h4 by hand)=  6.70131086
x2 to x3 (h5 computationally)=  1.55303335
x2 to x3 (h5 by hand)=  1.55303445
x2 to x3 (h5 by handdddd)=  1.553034
x3 to x1 (h6 computationally)=  4.37205346
x3 to x1 (h6 by hand)=  4.37205433
K22  (computationally) = 0.267691153
K22  (by hand) = 0.267691148
K32  (computationally) = 0.249329285
K32  (by hand) = 0.249329281
K21  (computationally) = 0.28627045
K21  (by hand) =0.286270449
K31  (computationally) = 0.284368602
K31  (by hand) = 0.284368894
K33  (computationally) = 0.23617235
K33  (by hand) = 0.23617316
K11  (computationally) = 0.492204778
K11  (by hand) = 0.49220651
x2= 0.125190815
x3= 0.142109715
x1=  0.266514817
T by hand=1099.99743
V by hand=-231.614045
m2=41.7
R=  0.0398218246
Energywithoutmass = 1539.60392
Energy=  33 731.6039
R=  0.0398218246
normx2 (computationally)=  -0.607324567
```

```
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.54499989
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.71768135
0 to x2 (h4 by hand)=  6.7176812
x2 to x3 (h5 computationally)=  1.55689486
x2 to x3 (h5 by hand)=  1.55689633
x2 to x3 (h5 by handdddd)=  1.55689551
x3 to x1 (h6 computationally)=  4.38324173
x3 to x1 (h6 by hand)=  4.38324251
K22  (computationally) = 0.268016156
K22  (by hand) = 0.268016114
K32  (computationally) = 0.249632677
K32  (by hand) = 0.249632648
K21  (computationally) = 0.286627516
K21  (by hand) =0.286627512
K31  (computationally) = 0.284723569
K31  (by hand) = 0.284723958
K33  (computationally) = 0.236460643
K33  (by hand) = 0.236461711
K11  (computationally) = 0.492824117
K11  (by hand) = 0.492825732
x2= 0.125038127
x3= 0.141935449
x1=  0.266176366
T by hand=1102.77621
V by hand=-231.902175
m2=41.8
R=  0.0398218246
Energywithoutmass = 1539.54064
Energy=  33731.5406
R=  0.0398218246
normx2 (computationally)=  -0.607324599
normx2 (by hand)=  -0.607324662
```

```
normx3 (computationally)=  -0.544999896
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622675
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.73405275
0 to x2 (h4 by hand)=  6.73405281
x2 to x3 (h5 computationally)=  1.56075548
x2 to x3 (h5 by hand)=  1.56075667
x2 to x3 (h5 by handdddd)=  1.56075613
x3 to x1 (h6 computationally)=  4.39443087
x3 to x1 (h6 by hand)=  4.3944316
K22  (computationally) = 0.268340943
K22  (by hand) = 0.268340962
K32  (computationally) = 0.249935784
K32  (by hand) = 0.249935797
K21  (computationally) = 0.286984162
K21  (by hand) =0.286984164
K31  (computationally) = 0.285077868
K31  (by hand) = 0.285078185
K33  (computationally) = 0.236747934
K33  (by hand) = 0.236748822
K11  (computationally) = 0.493442712
K11  (by hand) = 0.493444211
x2= 0.124885997
x3= 0.141761811
x1=  0.265839179
T by hand=1105.55511
V by hand=-232.189837
m2=41.9
R=  0.0398218246
Energywithoutmass = 1539.49393
Energy=  33731.4939
R=  0.0398218246
normx2 (computationally)=  -0.6073246
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999882
```

```
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622676
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.75042414
0 to x2 (h4 by hand)=  6.75042421
x2 to x3 (h5 computationally)=  1.5646164
x2 to x3 (h5 by hand)=  1.56461738
x2 to x3 (h5 by handdddd)=  1.56461705
x3 to x1 (h6 computationally)=  4.40562112
x3 to x1 (h6 by hand)=  4.405622
K22  (computationally) = 0.268665235
K22  (by hand) = 0.268665254
K32  (computationally) = 0.250238461
K32  (by hand) = 0.250238475
K21  (computationally) = 0.287340386
K21  (by hand) =0.287340388
K31  (computationally) = 0.285431779
K31  (by hand) = 0.285432044
K33  (computationally) = 0.237034917
K33  (by hand) = 0.237035658
K11  (computationally) = 0.494061092
K11  (by hand) = 0.494062853
x2= 0.124734415
x3= 0.141588803
x1=  0.265503253
T by hand=1108.33424
V by hand=-232.477289
m2=42.
R=  0.0398218246
Energywithoutmass = 1539.46356
Energy=  33 731.4636
R=  0.0398218246
normx2 (computationally)=  -0.607324495
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999892
normx3 (by hand)=  -0.544999908
```

```
normx1 (computationally)=  -0.434622679
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.76679474
0 to x2 (h4 by hand)=  6.7667941
x2 to x3 (h5 computationally)=  1.5684799
x2 to x3 (h5 by hand)=  1.56848223
x2 to x3 (h5 by handdddd)=  1.56848056
x3 to x1 (h6 computationally)=  4.41681189
x3 to x1 (h6 by hand)=  4.41681274
K22  (computationally) = 0.268988789
K22  (by hand) = 0.268988612
K32  (computationally) = 0.25054063
K32  (by hand) = 0.250540507
K21  (computationally) = 0.287696163
K21  (by hand) =0.287696146
K31  (computationally) = 0.285785832
K31  (by hand) = 0.285786441
K33  (computationally) = 0.237323127
K33  (by hand) = 0.237324765
K11  (computationally) = 0.494678466
K11  (by hand) = 0.49468029
x2= 0.124583373
x3= 0.14141644
x1=  0.265168576
T by hand=1111.11361
V by hand=-232.764626
m2=42.1
R=  0.0398218246
Energywithoutmass = 1539.44924
Energy=  33731.4492
R=  0.0398218246
normx2 (computationally)=  -0.607324543
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999895
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622681
```

```
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.78316736
0 to x2 (h4 by hand)=  6.78316704
x2 to x3 (h5 computationally)=  1.57234066
x2 to x3 (h5 by hand)=  1.57234248
x2 to x3 (h5 by handdddd)=  1.57234132
x3 to x1 (h6 computationally)=  4.42800355
x3 to x1 (h6 by hand)=  4.42800441
K22  (computationally) = 0.26931247
K22  (by hand) = 0.269312383
K32  (computationally) = 0.250842679
K32  (by hand) = 0.250842618
K21  (computationally) = 0.288051539
K21  (by hand) =0.28805153
K31  (computationally) = 0.286138786
K31  (by hand) = 0.286139267
K33  (computationally) = 0.237609172
K33  (by hand) = 0.237610483
K11  (computationally) = 0.495294988
K11  (by hand) = 0.495296792
x2= 0.124432886
x3= 0.141244681
x1=  0.264835136
T by hand=1113.893
V by hand=-233.051255
m2=42.2
R=  0.0398218246
Energywithoutmass = 1539.45121
Energy=  33731.4512
R=  0.0398218246
normx2 (computationally)=  -0.607324599
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999878
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.434622679
normx1 (by hand)=  -0.434622687
```

```
0 to x2 (h4 computationally)=  6.76843308
0 to x2 (h4 by hand)=  6.76843315
x2 to x3 (h5 computationally)=  1.56886386
x2 to x3 (h5 by hand)=  1.5688648
x2 to x3 (h5 by handdddd)=  1.56886451
x3 to x1 (h6 computationally)=  4.41793123
x3 to x1 (h6 by hand)=  4.41793221
K22  (computationally) = 0.269021509
K22  (by hand) = 0.269021529
K32  (computationally) = 0.250571
K32  (by hand) = 0.250571013
K21  (computationally) = 0.28773174
K21  (by hand) =0.287731742
K31  (computationally) = 0.285820657
K31  (by hand) = 0.28582091
K33  (computationally) = 0.237350402
K33  (by hand) = 0.237351109
K11  (computationally) = 0.494740328
K11  (by hand) = 0.494742274
x2= 0.124568307
x3= 0.141399221
x1=  0.265135175
T by hand=1111.39148
V by hand=-232.793098
m2=42.11
R=  0.0398218246
Energywithoutmass = 1539.44886
Energy=  33731.4489
R=  0.0398218246
normx2 (computationally)=  -0.607324551
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999882
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.43462268
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.77006975
```

```
0 to x2 (h4 by hand)=  6.77006949
x2 to x3 (h5 computationally)=  1.56925094
x2 to x3 (h5 by hand)=  1.56925248
x2 to x3 (h5 by handdddd)=  1.5692516
x3 to x1 (h6 computationally)=  4.41905032
x3 to x1 (h6 by hand)=  4.41905128
K22  (computationally) = 0.269053714
K22  (by hand) = 0.269053643
K32  (computationally) = 0.250601135
K32  (by hand) = 0.250601085
K21  (computationally) = 0.287767283
K21  (by hand) =0.287767277
K31  (computationally) = 0.285856203
K31  (by hand) = 0.285856608
K33  (computationally) = 0.237379656
K33  (by hand) = 0.237380758
K11  (computationally) = 0.494802011
K11  (by hand) = 0.494803961
x2= 0.124553236
x3= 0.14138203
x1=  0.265101789
T by hand=1111.66946
V by hand=-232.821892
m2=42.12
R=  0.0398218246
Energywithoutmass = 1539.44843
Energy=  33731.4484
R=  0.0398218246
normx2 (computationally)=  -0.607324541
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999888
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.43462268
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.77170683
0 to x2 (h4 by hand)=  6.77170651
```

```
x2 to x3 (h5 computationally)=  1.56963744
x2 to x3 (h5 by hand)=  1.56963919
x2 to x3 (h5 by handdddd)=  1.56963809
x3 to x1 (h6 computationally)=  4.42016937
x3 to x1 (h6 by hand)=  4.42017027
K22  (computationally) = 0.26908604
K22  (by hand) = 0.269085949
K32  (computationally) = 0.250631329
K32  (by hand) = 0.250631267
K21  (computationally) = 0.287802827
K21  (by hand) =0.287802818
K31  (computationally) = 0.285891606
K31  (by hand) = 0.285892067
K33  (computationally) = 0.237408545
K33  (by hand) = 0.237409798
K11  (computationally) = 0.494863559
K11  (by hand) = 0.49486542
x2= 0.124538173
x3= 0.141364842
x1=  0.265068414
T by hand=1111.94739
V by hand=-232.850585
m2=42.13
R=  0.0398218246
Energywithoutmass = 1539.4482
Energy=  33731.4482
R=  0.0398218246
normx2 (computationally)=  -0.607324518
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999874
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.43462268
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.77334379
0 to x2 (h4 by hand)=  6.7733433
x2 to x3 (h5 computationally)=  1.57002367
```

```
x2 to x3 (h5 by hand)=  1.57002547
x2 to x3 (h5 by handdddd)=  1.57002433
x3 to x1 (h6 computationally)=  4.42128874
x3 to x1 (h6 by hand)=  4.42128977
K22  (computationally) = 0.26911832
K22  (by hand) = 0.269118187
K32  (computationally) = 0.250661482
K32  (by hand) = 0.250661389
K21  (computationally) = 0.287838373
K21  (by hand) =0.28783836
K31  (computationally) = 0.285926949
K31  (by hand) = 0.285927421
K33  (computationally) = 0.237437219
K33  (by hand) = 0.23743849
K11  (computationally) = 0.494925593
K11  (by hand) = 0.49492772
x2= 0.124523114
x3= 0.141347657
x1=  0.265035055
T by hand=1112.22538
V by hand=-232.879354
m2=42.14
R=  0.0398218246
Energywithoutmass = 1539.44814
Energy=  33731.4481
R=  0.0398218246
normx2 (computationally)=  -0.607324494
normx2 (by hand)=  -0.607324662
normx3 (computationally)=  -0.544999891
normx3 (by hand)=  -0.544999908
normx1 (computationally)=  -0.43462268
normx1 (by hand)=  -0.434622687
0 to x2 (h4 computationally)=  6.77498074
0 to x2 (h4 by hand)=  6.7749801
x2 to x3 (h5 computationally)=  1.57041063
x2 to x3 (h5 by hand)=  1.57041294
```

x2 to x3 (h5 by handddddd)= 1.57041129

x3 to x1 (h6 computationally)= 4.42240766

x3 to x1 (h6 by hand)= 4.42240854

K22 (computationally) = 0.269150596

K22 (by hand) = 0.269150419

K32 (computationally) = 0.250691658

K32 (by hand) = 0.250691535

K21 (computationally) = 0.287873902

K21 (by hand) =0.287873885

K31 (computationally) = 0.285962448

K31 (by hand) = 0.285963053

K33 (computationally) = 0.237466415

K33 (by hand) = 0.237468041

K11 (computationally) = 0.494986913

K11 (by hand) = 0.494988797

x2= 0.12450806

x3= 0.141330485

x1= 0.265001703

T by hand=1112.50331

V by hand=-232.908047

m2=42.15

R= 0.0398218246

Energywithoutmass = 1539.4482

Energy= 33731.4482

R= 0.0398218246

normx2 (computationally)= -0.607324599

normx2 (by hand)= -0.607324662

normx3 (computationally)= -0.544999875

normx3 (by hand)= -0.544999908

normx1 (computationally)= -0.43462268

normx1 (by hand)= -0.434622687

0 to x2 (h4 computationally)= 6.77350839

0 to x2 (h4 by hand)= 6.77350846

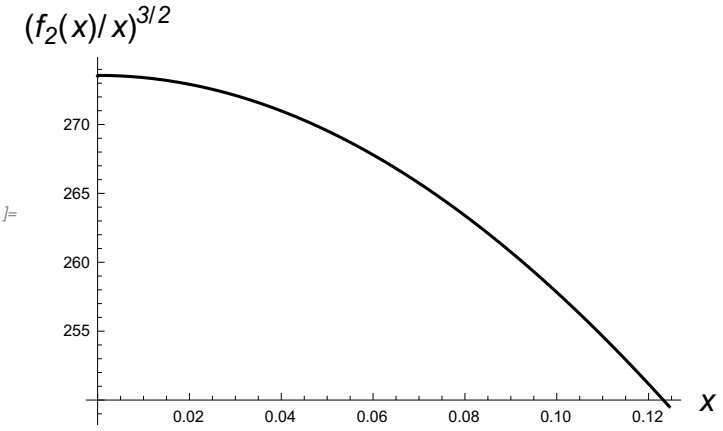
x2 to x3 (h5 computationally)= 1.57006087

x2 to x3 (h5 by hand)= 1.57006178

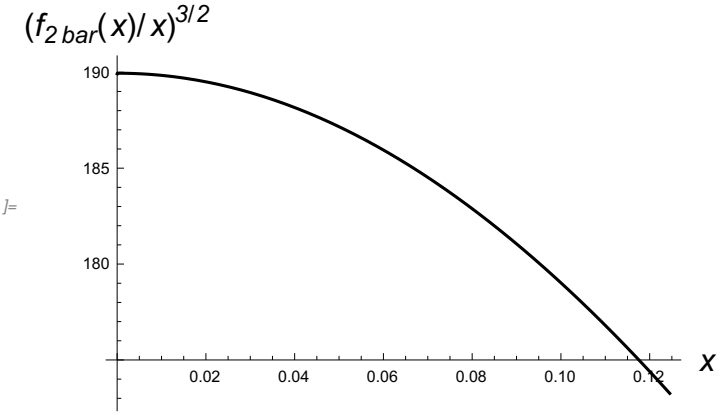
x2 to x3 (h5 by handddddd)= 1.57006153

```
x3 to x1 (h6 computationally)=  4.42140063
x3 to x1 (h6 by hand)=  4.42140165
K22  (computationally) = 0.269121829
K22  (by hand) = 0.269121847
K32  (computationally) = 0.250664634
K32  (by hand) = 0.250664647
K21  (computationally) = 0.287841938
K21  (by hand) =0.28784194
K31  (computationally) = 0.28593015
K31  (by hand) = 0.285930394
K33  (computationally) = 0.237439211
K33  (by hand) = 0.237439893
K11  (computationally) = 0.494931605
K11  (by hand) = 0.494933618
x2= 0.124521614
x3= 0.14134593
x1=  0.265031716
T by hand=1112.2531
V by hand=-232.882023
m2=42.141
R=  0.0398218246
Energywithoutmass = 1539.44824
Energy=  33731.4482
x1=0.265031716
x3=0.14134593
x2=0.124521614
Kinetic Energy=1112.2531
Potential Energy=-232.882023
Volume Energy=660.077161
```

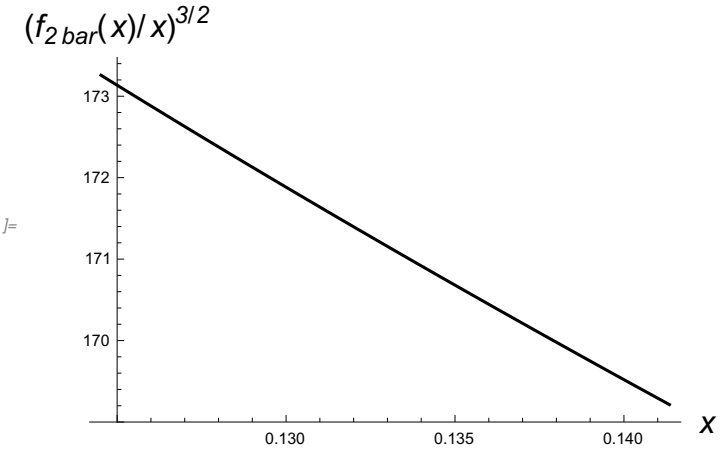
Out[ ]=


$(f_2(x)/x)^{3/2}$
270
265
260
255
0.02
0.04
0.06
0.08
0.10
0.12
x

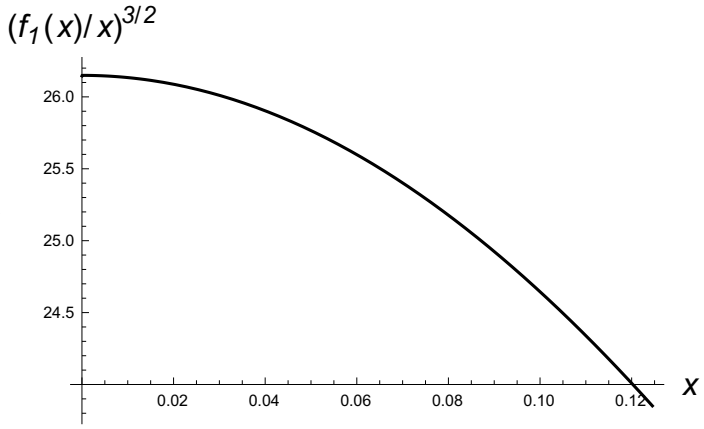
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
190
185
180
0.02
0.04
0.06
0.08
0.10
x

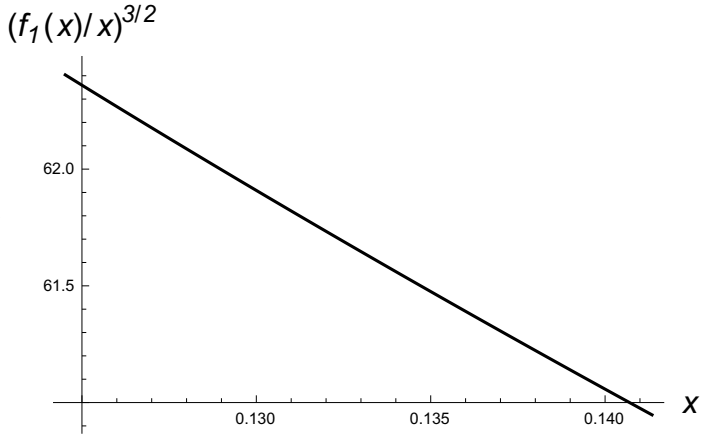
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
173
172
171
170
0.130
0.135
0.140
x

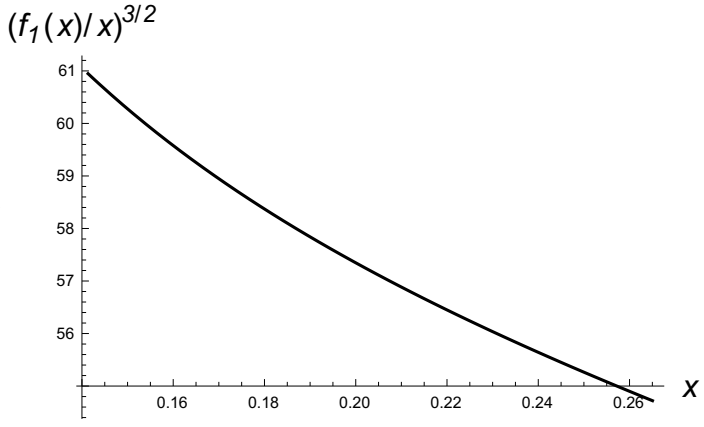
Out[ ]=


$(f_1(x)/x)^{3/2}$
26.0
25.5
25.0
24.5
0.02
0.04
0.06
0.08
0.10
0.12
x

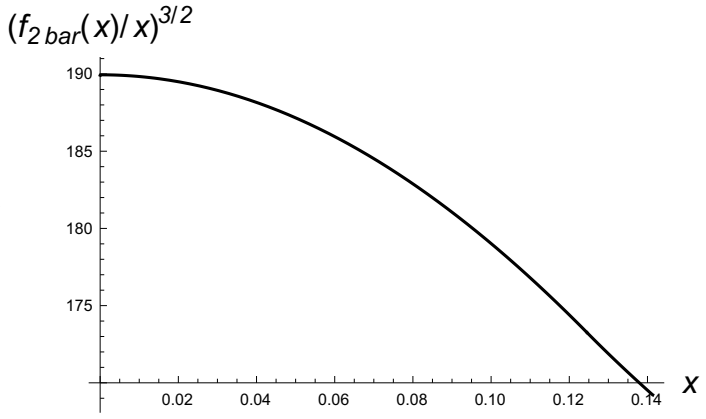
Out[ ]=


$(f_1(x)/x)^{3/2}$
62.0
61.5
0.130
0.135
0.140
x

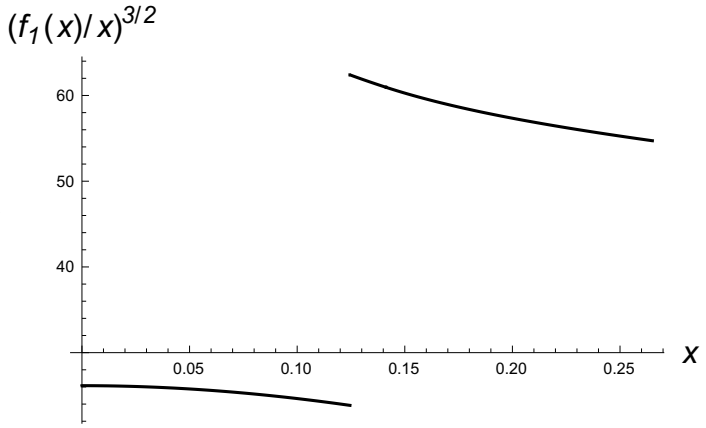
Out[ ]=


$(f_1(x)/x)^{3/2}$
61
60
59
58
57
56
0.16
0.18
0.20
0.22
0.24
0.26
x

Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
190
185
180
175
0.02
0.04
0.06
0.08
0.10
0.12
0.14
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
60
50
40
0.05
0.10
0.15
0.20
0.25
x